

Spectrum width

Y14 (2014) — English translation

To assess the degree of monochromaticity of the source emitting in the vicinity of the wave line $\lambda = 500 \text{ nm}$, a Young interference scheme was used (the distance between the slits in the opaque screen $d = 1 \text{ cm}$). The observation plane was moved into the Fraunhofer zone. Only bands whose intensity was at least 4 % the intensity of the zero band were observed visually. It turned out that when the width of the slits changes from $b_1 = 0.5 \text{ mm}$ to $b_2 = 0.1 \text{ mm}$, the number of observed bands increases from $N_1 = 40$ to $N_2 = 80$.

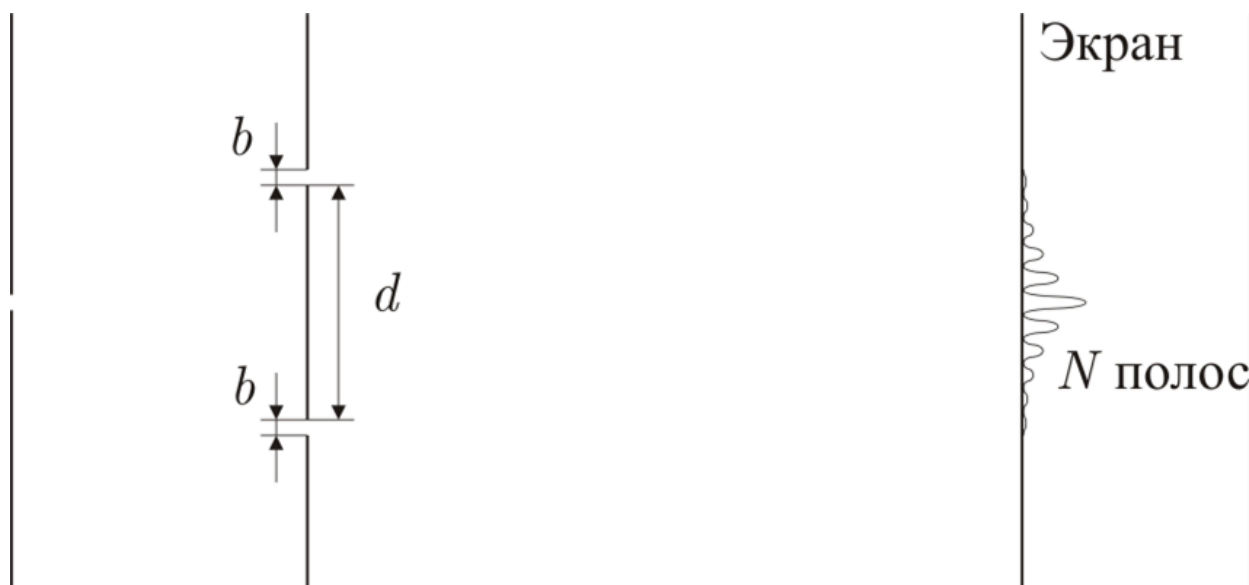


Fig. 1.

A1 Estimate the width of the emission spectrum $\Delta\lambda/\lambda$.

3.00

Source: <https://pho.rs/p/3993>